California at Irvine. These researchers noted that cosmic rays—elementary particles that stream through space and continually bombard our atmosphere—might also initiate production of microscopic black holes.

Cosmic ray particles were discovered in 1912 by the Austrian scientist Victor Hess; more than nine decades later, they still present many mysteries. Every second, cosmic rays slam into the atmosphere and initiate a cascade of billions of downward-raining particles that pass through your body and mine; some of them are detected by a variety of dedicated instruments worldwide. But no one is completely sure what kinds of particles constitute the impinging cosmic rays (although there is a growing consensus that they are protons), and despite the fact that some of these high-energy particles are believed to come from supernova explosions, no one has any idea of where the highest-energy cosmic ray particles originate. For example, on October 15, 1991, the Fly's Eye cosmic ray detector, in the Utah desert, measured a particle streaking across the sky with an energy equivalent to 30 billion proton masses. That's almost as much energy in a single subatomic particle as in a Mariano Rivera fastball, and is about 100 million times the size of the particle energies that will be produced by the Large Hadron Collider.[6] The puzzling thing is that no known astrophysical process could produce particles with such high energy; experimenters are gathering more data with more sensitive detectors in hopes of solving the mystery.

For Shapere and Feng, the origin of super-energetic cosmic ray particles was of secondary concern. They realized that regardless of where such particles come from, if gravity on microscopic scales is far stronger than formerly thought, the highest-energy cosmic ray particles might have just enough oomph to create tiny black holes when they violently slam into the upper atmosphere.

As with their production in atom smashers, such tiny black holes would pose absolutely no danger to the experimenters or the world at large. After their creation, they would quickly disintegrate, sending off a characteristic cascade of other, more ordinary particles. In fact, the microscopic black holes would be so short-lived that experimenters would not search for them directly; instead, they would look for evidence of black holes through detailed studies of the resulting particle showers